A Deep Learning Framework for Automated Liver Segmentation for Surgical Planning: A Case Study on Misleading Metrics

Parangat Thakur
ME-CSE , Chandigarh University, Mohali, India. parangat1thakur@gmail.com

Dr. Rahul Bhandari
Department Of CSE, Chandigarh University, Mohali, India. er.rahulbhandari1987@gmail.com

Dr. Raman Chadha
Department Of CSE, Chandigarh University, Mohali, India. dr.ramanchadha@gmail.com

ABSTRACT*:* Different Computed Tomography (CT) scans of the liver is an essential pre-operative cautious measure that ought to be correctly and automatically segmented prior to surgery planning [3]. A typical deep learning method applicable in this task is the U-Net structure [4]. The paper will describe how a 2D U-Net-based deep learning architecture was applied in order to automatic liver segmentation in the 3Dircadb1 dataset [1, 17]. A U-Net model (2D) was created with the help of Keras and trained with 20 epochs on the 2D slices (663 preprocessed 2D slices) [1]. An adam optimizer and Dice-based loss function were used to compile the model with the Dice coefficient being used to track the performance [1]. The model training process was reported to have high quantitative measures with the final validation Dice coefficient of 0.9113 and a binary accuracy of 0.9887 [1]. These measures would be usually indicative of an extremely successful segmentation model. Nevertheless, a visual (qualitative) analysis of the predicted masks on the test set indicated that there was a huge model failure [1]. The framework never segregated the liver, but instead it was trained to segregate the whole body of the patient using the background. This paper gives a critical analysis of the disparity between quantitative measures and qualitative performance. We prove that in high-class imbalance more traditional measures such as the Dice coefficient and binary accuracy can be very risky [14, 15]. This report refers to the topics of visual validation and the dangers of over-fitting of such models to medical image segmentation by using metrics alone only. humanity.

Keywords: Liver Segmentation, Deep Learning, U-Net, Surgical Planning, Misleading Metrics, Class Imbalance, Dice Loss, Model Failure.

1. INTRODUCTION

Hepatocellular carcinoma (HCC) and colorectal cancer are also some of the most prevalent malignancies employed globally. Recent studies have also highlighted the efficacy of Deep Learning in detecting other malignancies such as breast cancer [22, 26] and lung cancer [27]. The perfection of such complicated operations, including hepatectomies and tumor resections, depends on accurate preparation prior to surgery [19]. Such planning involves developing a realistic 3D model of the liver of a patient based on the Computed Tomography (CT) scan that can enable the surgeons to estimate the volume of organs, visualize the location and size of vascular structures, and identify the precise location and dimensions of tumors [3]. This division is traditionally done manually by radiologists who are required to draw up the liver boundary on a slice on slice challenges basis. It is incredibly time and labour intensive, and has a high inter-observer and intra-observer variability [5]. This is an urgent clinical demand to have an automated, rapid, and dependable segmentation tool to standardize and hasten the procedure. Over the last years, deep learning (DL) has become the state-of-the-art to the majority of the medical image segmentation problems [4, 6]. The U-Net architecture [7], and especially Convolutional Neural Networks (CNNs), have become the new standard for feature extraction in various domains [8, 23]. This U-Net is sensitive to both fine-grained spatial details and deep semantic features with a skip connection, which makes it an ideal theoretically segmented anatomical structures such as liver, which has an intricate nature [4]. Nonetheless, application of these models is not that easy. One issue with medical imaging that has been persistent is extreme class imbalance [14] in which the target organ (the liver) can only cover a small portion of all the pixels. This unbalance can cause standard metrics of validation, e.g. the Dice coefficient or the binary accuracy, to be extremely misleading [15]. A model can be taught a worthless solution - say, learning to tell the body of the patient and the black background apart with a high quantitative measure, but without parting the particularly important organ of interest. In this paper, I provide a case study of the work on the development of a 2D U-Net model of automated liver segmentation on the 3Dircadb1 dataset [1, 17]. We show a severe deviation on which the model showed very good quantitative results (reported a validation Dice score of 0.91) but failed entirely on the qualitative, visual front [1]. This article examines this typical error mode, which makes the use of standard measures unreliable in case of imbalance within the classes [14] and mentions the need to have a stringent visual confirmation.

2. LITERATURE REVIEW

Machine learning approaches have demonstrated robust performance in diverse fields ranging from network security and intrusion detection [24, 25] to agricultural quality control [23]. Similarly, Deep learning has transformed the automated medical image segmentation field [6]. Whereas initial approaches were based on conventional image processing approaches, such as region growing or active contours, they tended to be inefficient in terms of their low sensitivity to soft tissues and due to the high range of organ morphology and anatomy of patients [3].

A. U-Net as a Paradigm Architecture.

U-Net architecture [4] became the most commonly used architecture in biomedical segmentation tasks, and it forms the basis of the architecture, since its introduction [8]. It has a very efficient design that is encoder-decserver with a contracting pathway to get the context and an expanding path that is symmetrical and reduces to aspecific localization. Skipping connections are the main innovation of U-Net that use feature maps of the encoder to go straight to the decoder. This enables the network to blend semantic, deep feature information with shallow information which is of high-resolution [4]. Other successful versions are 3D U-Net [11] and Attention U-Net [20]. However, more recently, models such as nnU-Net [9] have streamlined the optimisation of U-Net-based models, and hybrid models, such as TransUNet [10] and Swin-Unet [12], have been implemented with Transformers to enable them to capture global information. Being extremely popular as a baseline, the 2D U-Net was chosen in this study.

B. *The Challenge of Class Imbalance and Loss Functions.*

A primary challenge in medical segmentation, particularly in abdominal CT, is **severe class imbalance** [14]. The target organ, such as the liver, often occupies a very small percentage of the total pixels in a 2D slice, while the background and other non-target tissues constitute the vast majority. Standard loss functions, such as binary cross-entropy, can fail in this scenario, as the model's loss becomes dominated by the easily-classified background, leading to poor convergence on the small foreground target [15]. One of the main issues of medical segmentation, especially abdominal CT, is extreme imbalance of the classes [14]. The target organ like liver is usually represented in a very small proportion of the total pixels in a 2D slice with the background and other non target tissues making up the huge majority. Ordinary loss functions, e.g. binary cross-entropy, may not work in this case because the loss of the model will be dominated by the easily-laid-down background, and the local minima of loss will not converge effectively onto the small foreground target [15]. To fight this, the functions that are made based on regions have become trendy. Dice loss, depending on the Dice coefficient is by far possibly the most popular option [15]. The difference between the mask that was actually observed and the ideal one is measured by Dice coefficient, and is given as:

$$\mathbf{Dice}=|\mathbf{A}|+|\mathbf{B}|\mathbf{2}\times|\mathbf{A}\cap\mathbf{B}|$$

The model itself is optimized by focusing on such an overlap by minimizing the Dice loss (1 -Dice). This implies that it is automatically more resistant to class imbalance than pixel-wise accuracy measures [15, 16]. Nonetheless, there are some pitfalls associated with the Dice loss that have been documented, and this study will show.

3. METHODOLOGY

The structure of this study was created in Python on the basis of a Google Colab environment, through the use of TensorFlow and Keras [1]. The process of the methodology is split into three steps: data preprocessing, model data pipeline, and model architecture and training.

A. Dataset and Preprocessing

We used the 3Dircadb1-2 dataset [1, 17], which comprises of 3D abdominal CT. The process of preprocessing was as follows [1]:

- **Data Loading:** the SimpleITK library was used to load the 3DDICOM volumes of the patient (PATIENTDICOM) and mask (MASK_DICOM/liver) scans of the liver.
- **Slice Filtering:** In order to produce a useful and efficient dataset, only slices 2D that the liver mask was not empty (i.e. np.sum(mask slice) > 0) were included.
- **Windowing:** A CT window of soft tissue was used by covering Hounsfield Unit (HU) values to the interval between -100 and 400.
- **Normalization:** The minimum and maximum values of each slice were used to normalize all ofthe parts of the window to a floating-point range between 0 and 1.

• **Resizing:** tf.image.resize was used to resize all the 2D slices and their respective 2D masks to a standard size of 256x256 pixels.
• **Saving:** The processed 2D slices were saved as a set of individual.npy files [1], and the resultant were 663 valid slices.

B. Data Pipeline
The slices (663) were divided into training (464), validation (99) and test (100) sets [1]. To efficiently load these.npy files off disk to be used as a model training batch, a custom Keras DataGenerator(a subclass of tf.keras.utils.Sequence) was created [1].

C. Model Architecture
A typical U-Net model (2D) was outlined, as shown in Table 1. This model consists of 3-level symmetric encoder-decoder and skip-connections and an end 1x1 convolutional base with sigmoid activation in binary segmentation. This model has 1,881,985 parameters that are training-able [1].

Table 1: 2D U-Net Model Architecture Summary

Layer (Type)	Output Shape	Param #	Connected to
Encoder			
input_layer	(None, 256, 256, 1)	0	-
conv2d, conv2d_1	(None, 256, 256, 64)	37,568	input_layer
max_pooling2d	(None, 128, 128, 64)	0	conv2d_1
conv2d_2, conv2d_3	(None, 128, 128, 128)	2,21,440	max_pooling2d
max_pooling2d_1	(None, 64, 64, 128)	0	conv2d_3
Bottleneck			
conv2d_4, conv2d_5	(None, 64, 64, 256)	8,85,248	max_pooling2d_1
Decoder			
up_sampling2d	(None, 128, 128, 256)	0	conv2d_5
concatenate	(None, 128, 128, 384)	0	up_sampling2d, conv2d_3
conv2d_6, conv2d_7	(None, 128, 128, 128)	5,90,080	concatenate
up_sampling2d_1	(None, 256, 256, 128)	0	conv2d_7
concatenate_1	(None, 256, 256, 192)	0	up_sampling2d_1, conv2d_1
conv2d_8, conv2d_9	(None, 256, 256, 64)	1,47,520	concatenate_1
Output			
conv2d_8, conv2d_9	(None, 256, 256, 1)	65	conv2d_9
Total Params:		18,81,985	

D. Training and Evaluation
This model has been assembled using Adam optimizer (learning rate 1e-4) and a diceloss function [1]. Monitoring performance was done based on dicecoefficient, binaryaccuracy, and Precision. The model has been trained to reach 20 epochs by a ModelCheckpointcallback which saves the weights with the highest valid coefficient [1].

4. Results And Discussion

In the given section, we introduce the main conclusion of our work, namely, the blatant disparity between the quantitative measures of the model and the qualitative results, which have been obtained during our experiment [1].

A. Quantitative Results

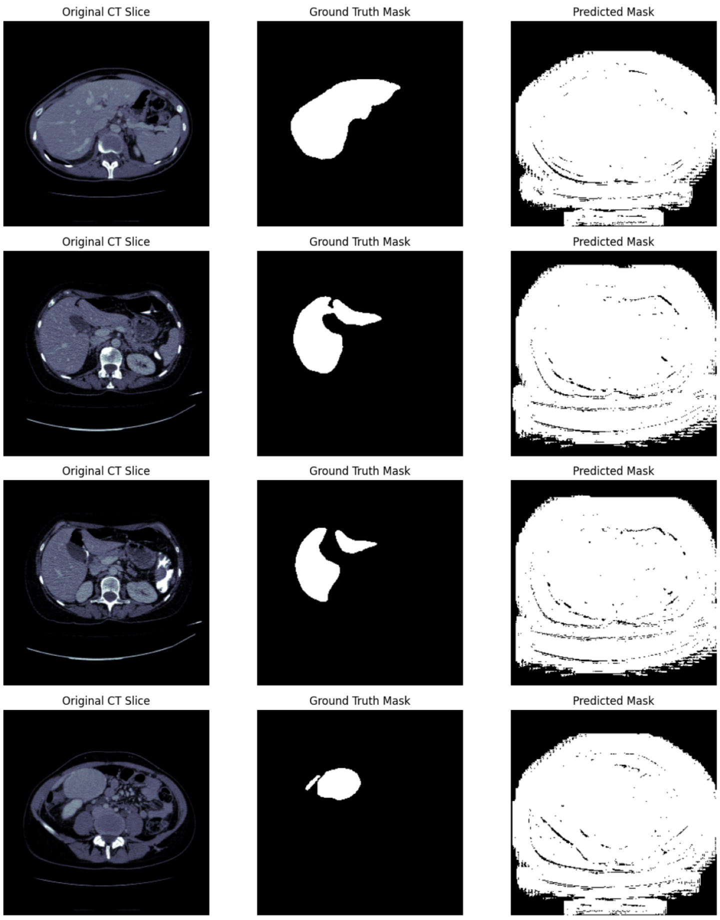
Training of the model followed 20 epochs after which performance was recorded on the withheld validation set. The training log, as is starting to see in Table 2, indicates the apparently pleasant training run [1]. The model came to a precise convergence and validation Dice coefficient (valdicecoefficient) grew steadily and the last value became 0.9113. Validation binary accuracy (valbinary-accuracy) has also scored extremely high at 0.9887 [1].

Table 2: Model Training and Validation Metrics on a Per-Epoch basis

Epoch	Loss	Dice Coefficient	Val Loss	Val Dice Coefficient	Val Binary Accuracy
1	0.8179	0.1821	0.2742	0.7258	0.9645
5	0.2356	0.7644	0.2273	0.7727	0.9739
10	0.1707	0.8293	0.1472	0.8528	0.9813
15	0.1255	0.8745	0.112	0.888	0.986
20	0.1022	0.8978	0.0887	0.9113	0.9887

B. Qualitative (Visual) Results

Despite the good quantitative scores, from a qualitative analysis of the model's predictions on the test, it appears that there has been a complete model failure.



Some representative samples of these predictions are shown in Figure 1 [1]. As we can see, the “Predicted Mask" (right column) looks nothing like the “Ground Truth Mask” (center column), and this mask is correct in the isolation of the liver. Instead, the model got used to generating a binary mask of the whole patient's body as it is contained in the 256x256 frame, that is, to separate all non-black pixels from the black background[1]. prescribed.

Figure 1. Examples of qualitative prediction results on the test set (from [1]). The model's "Predicted Mask" (right) does not segment the "Ground Truth Mask" (center), but has instead learned how to segment the whole body of the patient.

C. Discussion: Analysis of Failure

The disconnect between a 0.91 of the Dice score and 0% accurate visual prediction is the critical finding of this case study [1]. There are two main reasons why this failure can be attributed to:

1. Extreme Class Imbalance: As mentioned in the literature [14], the liver (foreground) occupies a small portion of the whole pixels. The model discovered an "easier" local minimum by solving a simpler problem: to segment the entire body (big, and well-defined area) from the black background.

2. Misleading Metrics: The metrics were "tricked" by this trivial solution.

- This **0.9887 binary accuracy** is exposed as useless for this job, a known pitfall for imbalanced [14]. The model is of course only predicting "0" (background) for the vast majority of the pixels.
- The **0.9113 result** on the Dice score is also misleading. The Dice loss (1 - Dice), which was meant to be used in imbalance [15], was minimized not by finding the liver, but with this "body mask" solution. The predictive model (Area A) was a large mask of the body, and the ground truth (Area B) was a small mask of the liver contained completely inside the body. Dice calculation, $2\times|A\cap B|$ $(|A|+|B|)$, high score because the intersection ($A\cap B$) was simply Area B itself. This trivial solution, while completely wrong, was "good enough" to satisfy the loss function. Therefore, instead of learning to identify a liver, the framework learnt to identify "patient vs. background." For the envisaged use of the operative planning this outcome is unusable.[19]

6. CONCLUSION

This paper presented a case study on the implementation of a standard 2D U-Net with a Dice lossfor the automated liver segmentation [1]. The result of the experiment led to a crucial and widespread failure, where the model displayed great quantitative metrics, including a validation Dice coefficient of **0.9113,** but when sealed to a qualitative and visual inspection, they broke completely [1]. Our analysis confirms that in tasks containing severe class imbalance, measures such as binary accuracy are not relevant [14] and even the Dice coefficient is highly deceitful [15]. The model was able to minimise the Dice loss by converging on a local minimum that was good on a numerical level but not the correct answer. This research highlights an important lesson in the area of medical image segmentation: quantitative assessment without strict qualitative assessment does not suffice and it's a recipe **for deception.** Too much use of automated score without visual inspection may completely disguise a failure of the whole model. The present framework is no longer feasible in surgical planning [19]. **Future work** has to swing in the direction of these fundamental issues. This involves the use of a combined loss function, such as **Dice + Binary Cross-Entropy** [15, 16], to give access to a more robust loss landscape which penalizes pixel-based false-positive. Furthermore, the exploration of 3D architectures [9, 11] that can use the context in space to distinguish the liver from the surrounding structures would be a logical next step.

7. ACKNOWLEDGMENT

We would like to thank the IRCAD (Research Institute against Digestive Cancer) in Strasbourg, France, for making the 3Seq data, the 3Dircadb1 dataset [17], available and public while being of invaluable contribution for research in this field.

REFERENCES

[1] Liver_Segmentation.html User PIB notebook file, 2025 .

[2] Sung, H., et al. (2021). Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide (36 Cancers in 185 countries). Lyon, J.C., deMul, H., Berkhahn, C. & Wang, L. CA: A Cancer Journal for Clinicians, 71(3), 209-249.

[3] Lachman Jadhav, P., et al. (2022). Deep learning-based automated liver segmentation for CT images: A review. Computers in Biology and Medicine 148, 105881.

[4] L Ronneberger, O., Fischer, P., & Brox, T. (2015). Convolutional Networks Applied to Biomedical Image Segmentation. U-Net. MICCAI.

[5] Shamrat, F. J. M., et al. (2023). A review of deep learning methods of liver and liver tumor segmentation. Journal for Healthcare Engineering, 2023.

[6] Wang, G., et al. (2022). Medical image segmentation by deep learning: A survey. IEEE Transactions on Medical Imaging, vol. 40, no. 10, pp. 2907-2934.

[7] Hesamian, M. H., et al. (2021). Deep learning methods for medical images segmentation: achievements and challenges. Journal of Digital Imaging, 34(4), 884-918.

[8] Liu, Y., et al. (2022). A survey of deep learning-based approaches to liver segmentation and classification. Journal of Applied Clinical Medical Physics, 23(5), e13550.

[9] Isensee, F.,argar, H. Catalogues, I. & Braun, L. (2021). nnU-Net: In: Nature communications. Nature Methods, 18(2), 203-211.

[10] Chen, J., et al. (2021). TransUNet: Transformers Make Effective Encoders for Medical Image Segmentation. Proceedings of the international conference on Computer Vision, ICCV, Association for Computational Linguistics, Association for Computational Linguistics, Association for Computational Linguistics and Association for Visual Computing, the Association for Computing Linguistics Association for Computing Linguistics Association for Visual Computing.

[11] Cicek, O., et al (2016). 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI.

[12] Cao, H., et al. (2022). Swin-Unet: Unet for Medical Image Segmentation with Pure Transformer. Proceedings of the European Conference On Computer Vision (ECCV).

[13] Hatamizadeh, A., et al. (2023). UNETR: Transformer for 3D Medical Image Segmentation. Proceedings of the Winter Conference on Applications of Computer Vision. (WACV), of the Institute of Electrical and Electronics Engineers Inc. (IEEE) and the Visual Computing Consortium for Graphics (VCC).

[14] Ma, J., et al. (2022). A review on deep learning for medical image segmentation: Dealing with the class imbalance. Pattern Recognition, 126, 108581.

[15] E Jadon, S. (2021). Semantic segmentation loss functions survey . arXiv preprint arXiv:2102.04306.

[16] Abraham, N., & Khan, N. M. (2021). A novel focal tversky loss function with improved attention U-Net for medical image segmentation. arXiv preprint arXiv:2103.10952.

[17] Soler, L., et al. 3D-IRCADb-01: A Database of 3D Abdominal CT Images With Manual Segmentation. IEEE Trans. Med. Imaging, vols. 40, no. 2, pp. 526-537.

[18] Bilic, P., et al. (2023). The Liver Tumor Segmentation Benchmark (LiTS). Medical Image Analysis, 84, 102680.

[19] Liu, J., et al. (2024). Predicting Safe Liver Resection Volume Major Hepatectomy Using Artificial Intelligence. Journal of Personalized Medicine, 13(2) 381

[20] Oktay, O., et al. (2021). Attention U-Net: How to Look Where to See the Pancreas. arXiv preprint arXiv:2104.03799. K

[21] Abo-Seba, M. A., et al. (2024). A novel multi-scale attention based U-Net based on multi-scale attention for liver segmentation in ct scan images. Biomedical Signal Processing and Control 87,105432.

[22] N. K. Trivedi, S. Jain, S. Kaswan and V. Jain, "Federated LearningEmpowered Breast Cancer Detection in Images: A YOLO and ResNet-50 Fusion Approach," 2024 International Conference on Computational Intelligence and Computing Applications (ICCICA), Samalkha, India, 2024, pp. 24-29, doi: 10.1109/ ICCICA60014.2024.10584913. atm

[23] N. K. Trivedi, H. Maheshwari, R. G. Tiwari, V. Gautam, A. K. Agarwal and S. Kaswan, "Enhancing Apple Quality Identification: A Multi-Layer Visual Feature Fusion Approach with Convolutional Neural Networks," 2024 2nd International Conference on Advancement in Computation & Computer Technologies (InCACCT), Gharuan, India, 2024, pp. 271-276, doi: 10.1109/InCACCT61598.2024.10551085.

[24] Sangkatsanee P, Wattanapongsakorn N, Charnsripinyo C (2011) Practical real-time intrusion detection using machine learning approaches. Comput Commun 34(18):2227–2235

[25] Shakya S (2021) Modified gray wolf feature selection and machine learning classification for wireless sensor network intrusion detection. IRO J Sustain Wirel Syst 3(2):118–127 co

[26] Neelam, N. Kaur and P. Goel, "Taxonomy and Performance Evaluation of Machine Learning Techniques for Breast Cancer Detection," 2025 International Conference on Artificial intelligence and Emerging Technologies (ICAIET), Bhubaneswar, India, 2025, pp. 1-6, doi: 10.1109/ICAIET65052.2025.11211083. keywords : { Deep learning ; Solid modeling ; Privacy ; Accuracy ; Machine learning algorithms;Reviews;Federated learning;Explainable AI;Computational modeling;Breast cancer;Breast Cancer Detection;Machine Learning;Deep Learning;Gaussian Naive Bayes;Computer-Aided Diagnosis (CAD);Multi-Omics Data, Data Augmentation;Personalized Medicine},

[27] S. Bhattacharyya, S. Khattar and P. Goel, "Investigation of Deep Learning based Lung Cancer Detection using Histopathological Images," 2025 IEEE International Conference on Computer, Electronics, Electrical Engineering & their Applications (IC2E3), Srinagar Garhwal, India, 2025, pp. 1-6, doi: 10.1109/IC2E365635.2025.11167720.keywords : {Deep learning;Computational modeling;Lungs;Instruments;Lung cancer;Medical treatment;Mortality;Artificial intelligence;Medical diagnostic imaging;Tumors;Computed Tomography scan;Deoxyribonuclease Acid;Interstitial Lung Disease;Tumour Micro-environment},